

1. Correlations and Partial correlations

Correlation Matrix The correlation matrix is basic to many kinds of analysis. It is a bridge over which scientists can move from their data to sophisticated statistical analyses of patterns, dimensions, factors, dependencies, discriminations or hierarchies. In its own right, however, the correlation matrix contains much useful knowledge.

1. Each coefficient measures the degree and direction (sign) of the correlation (linear relationship) between the row and column variables.
2. Each correlation squared defines the proportion of covariation between these variables.
3. Each correlation is the cosine of the angle between the variables as vectors of mean-deviation data.

The discussion above is referred to Pearson's correlation coefficient between two variables. The Spearman correlation coefficient is a coefficient between the ranks of the variables. Traditionally it has been used with ordinal variables, but when applied to continuous variables it informs you about the monotonicity of the relationship between the variables, i.e., as X gets larger, Y keeps getting larger, or keeps getting smaller. It doesn't refer to linearity, only tells you if the underlying relationship between the variables is monotonic. Its value is again between -1 and 1 and a value close to 0 means the ranks of one variable do not covary with the ranks of the other variable; in other words, as the ranks of one variable increase, the ranks of the other variable do not increase (or decrease).

A possible interpretation of the magnitude (in absolute value) of Pearson's correlation coefficient between two variables is the following:

- $1 - 0.9$, very high: very strong linear relationship, variables are measuring almost the same thing.
- $0.9 - 0.6$, high: strong relationship, you can be confident that these two variables are connected in some way.
- $0.6 - 0.3$, moderate: moderate relationship, you might be on to something or you might not. If it is close to 0.3 it is better to say it is low or weak, not moderate.
- $0.3 - 0$, negligible: very weak relationship, it may be an artifact of the data set and in fact there is no linear relationship at all.

Partial Correlations Partial correlations are used to explore the relationships between pairs of variables after we take into account the values of other variables (*holding constant* or *independent* of the other variables). Partial correlations should be compared to the corresponding ordinary correlations. When interpreting partial correlations, three results can potentially occur. Each of these results yields a different interpretation.

1. Partial and ordinary correlations are approximately equal. This suggests that the relationship between the variables of interest cannot be explained by the remaining explanatory variables upon which we are conditioning.

2. Partial correlations closer to zero than ordinary correlations. Many times, this is the type of case that we are looking for. In this case, this would suggest that the relationship between the variables of interest might be explained by their common relationships to the explanatory variables upon which we are conditioning. For example, we might find the ordinary correlation between blood pressure and blood cholesterol might be a high, strong positive correlation. We could potentially find a very small partial correlation between these two variables, after we have taken into account the age of the subject. If this were the case, this might suggest that both variables are related to age, and the observed correlation is only due to their common relationship to age.
3. Partial correlations farther from zero than ordinary correlations. This rarely happens. This situation would suggest that unless we take into account the explanatory variables upon which we are conditioning, the relationship between the variables of interest is hidden or masked.